Lab 0 — Computers Speak Vectors

Friday, August 28

Last updated: August 18, 2026.

In class Friday, August 28 (same day as Lecture 03). Due date is on the schedule. Submit in **Canvas**.

Pick **Octave or Python** and stay with that language this term. No prior coding is assumed. Bring a laptop, or use a campus lab machine.

Software: GNU Octave or Python with NumPy and Matplotlib. See Resources.

Indexing (read this first)

Boyd writes x_1 for the first entry. Octave counts from **1**. Python/NumPy count from **0**.

Meaning	Boyd	Octave	Python
first entry of x	x_1	`x(1)`	`x[0]`
second entry	x_2	`x(2)`	`x[1]`
length n	n	`length(x)`	`len(x)`

What to submit

Upload one file (`.m`, `.py`, or a PDF/screenshot of your session) **and** type short written answers in the Canvas text box, or include them in the same document.

You should have:

1. Code that creates the vector and prints the requested entries.
2. A sentence about the two modified entries.
3. A plot of x, $2x$, and $-x$ (screenshot or saved image) and two sentences on the geometry.

The data

Use this 6-vector of **Arcata morning temperatures**, in °F, for six successive hours:

$$x = (52, 51, 53, 55, 58, 60).$$

Entry x_1 is 7 a.m.; x_6 is noon.

C0.1 — Enter and index

Create x in software. Print n, the first entry, the last entry, and x_3.

Octave:

```
x = [52, 51, 53, 55, 58, 60];
n = length(x)
x(1)
x(n)
x(3)
```

Python:

```python
import numpy as np
x = np.array([52, 51, 53, 55, 58, 60])
n = len(x)
print(n, x[0], x[-1], x[2])
```

Written: In one sentence, what quantity does x_3 represent, including units?

C0.2 — Modify two entries

The 9 a.m. reading was mistyped: it should be 54 instead of 53. The noon reading should be 61 instead of 60. Change those two entries. Print the new vector.

Written: Which Boyd indices did you change, and what changed in the weather story?

C0.3 — Plot x, $2x$, and $-x$

Plot the original x, the scaled vector $2x$, and the vector $-x$. Use hour index $i = 1, \ldots, 6$ on the horizontal axis (or hours 7 a.m. to noon). Include a legend.

Octave:

```
i = 1:6;
plot(i, x, '-o', i, 2*x, '-o', i, -x, '-o');
legend('x','2x','-x');
xlabel('hour index i');
ylabel('value');
```

Python:

```python
import matplotlib.pyplot as plt
i = np.arange(1, 7)
plt.plot(i, x, '-o', label='x')
plt.plot(i, 2*x, '-o', label='2x')
plt.plot(i, -x, '-o', label='-x')
plt.xlabel('hour index i')
plt.ylabel('value')
plt.legend()
plt.show()
```

Written: In two sentences, what did multiplying by 2 do geometrically, and what did multiplying by -1 do? (Temperature units on $-x$ will look odd; that is the point—the picture is about the vector operation, not a new weather forecast.)

If something breaks

Copy the error message into your Canvas submission and say which line you changed. We would rather see a debug note than a blank upload.